A Note on Infinite Horizon Alternating Offers Bargaining

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1 Introduction

We study the bargaining game [claim:thm-1] [1]. The formal derivation is presented in Appendix A.

2 Main Results

Theorem 1. *The unique subgame perfect equilibrium payoff to Player 1 is* $x^* = \frac{1-\delta_2}{1-\delta_1\delta_2}$.

3 Appendix: Mathematical Derivations

Let M_1 and m_1 be the supremum and infimum of Player 1's SPE payoffs. In round 2, Player 2 accepts any offer $y > \delta_2 M_1$, yielding $m_1 \geq 1 - \delta_2 M_2$. Symmetrically, $M_1 \leq 1 - \delta_2 m_2$. Solving yields $M_1 = m_1 = \frac{1-\delta_2}{1-\delta_1\delta_2}$.

References

[1] Ariel Rubinstein. Perfect equilibrium in a bargaining model. *Econometrica*, 1982.